DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
UNIVERSITY COLLEGE OF ENGINEERING(A)
Osmania University, Hyderabad-07

CLASS TEST-I

SUBJECT: Artificial Intelligence and Machine Learning
Class: BE (CSE)
Semester: V

Date: 07-09-2026
Max. Marks: 30
Duration: 1.5 Hours (9.30 – 11:00 am)

SNo			Marks	BT	CO
		PART-A **Answer All Questions(5Q X 2M =10M)**			
1.	a)	Differentiate between Strong AI and Weak AI with one suitable example each?	2	2	1
	b)	What is a state space graph? How is it useful in solving an AI problem?	2	2	2
	c)	What is a game tree? How is it used in the Minimax algorithm?	2	2	2
	d)	What is propositional Logic? Give one example of a proposition used for knowledge representation.	2	1	1
	e)	Differentiate between frames and semantic networks as knowledge techniques.	2	2	1
		PART-B **Answer any Two Questions(2Q X 10M =20M)**			
2.	a)	Explain the Turing Test and the major approaches to Artificial intelligence. Describe the applications of AI in different domains with suitable examples.	5	2	1
	b)	Using a suitable example, demonstrate the Generate-and-Test approach and show how the corresponding problem can be modeled as a state-space graph	5	3	2
3.	a)	Explain Best First Search and Beam Search. Discuss the role of Heuristic functions in informed search and analyse the advantages and limitations of these search strategies.	5	4	2
	b)	Explain the different knowledge representation techniques used in Artificial Intelligence. Discuss Semantic Networks, Frames, and Scripts with suitable examples.	5	2	1
4.	a)	Explain Alpha Beta pruning with a suitable game tree. Demonstrate how pruning reduces the number of nodes evaluated while obtaining the same optimal decision as Minimax.	5	3	3
	b)	Explain Propositional Logic and Predicate logic as knowledge representation techniques. Illustrate the representation of a simple real world problem using suitable logical statements.	5	3	1